



SHREYARTH UNIVERSITY

School of Computer Science and Application
BCA

Course Name: Introduction to Operating System

Course Code:

Semester: 4th

Assignment 2

Deadline: Till 22-03-2025

Module- 2: Deadlock & Device Management

Sr No.	Questions																																																																						
1.	Apply banker’s algorithm to the following snapshot taken at time t0 and find out process execution sequence that lands a system in safe state. <table><tr><th></th><th colspan="3">Allocation</th><th colspan="3">Max</th><th colspan="3">Available</th></tr><tr><th></th><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th></tr><tr><td>P₀</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P₁</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₂</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₃</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₄</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></table>		Allocation			Max			Available				A	B	C	A	B	C	A	B	C	P ₀	0	1	0	7	5	3	3	3	2	P ₁	2	0	0	3	2	2				P ₂	3	0	2	9	0	2				P ₃	2	1	1	2	2	2				P ₄	0	0	2	4	3	3			
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2.	Suppose Disk drive has 300 cylinders. The current position of head is 70. The queue of pending request is 42,85,25,130,230,270,85,190 Calculate head movement for the following algorithms. 1. FCFS 2. SSTF 3. SCAN																																																																						
3.	Explain Deadlock Prevention.																																																																						
4.	Explain Secondary storage Structure.																																																																						
5.	Explain RAID levels in details.																																																																						

Prepared By: - Prof.Miki Patel